

# ESP32-NUT LCD Integration Guide

## Adding Elecrow CrowPanel 2.4" Display

Instructions to add an Elecrow CrowPanel Advance 2.4" HMI display (320×240, model DIS01624A) to the esp32-nut ESP32-S3 UPS monitoring firmware, showing a live dashboard of UPS telemetry.

### 1. Add LVGL Configuration

Create **include/lv\_conf.h** with LVGL 8.3.11 settings:

- Download template from:

[https://raw.githubusercontent.com/lvgl/lvgl/v8.3.11/lv\\_conf\\_template.h](https://raw.githubusercontent.com/lvgl/lvgl/v8.3.11/lv_conf_template.h)

- Enable content: **#if 0** → **#if 1**
- Enable fonts: **LV\_FONT\_MONTERRAT\_20**, **LV\_FONT\_MONTERRAT\_32**
- Set **LV\_TICK\_CUSTOM 1**
- Disable examples: **LV\_BUILD\_EXAMPLES 0**

### 2. Create LcdDisplay Module

Create **lib/LcdDisplay/src/** with three files:

#### **lgfx\_config.h**

LovyanGFX board driver configuration (ST7789 SPI, FT6336 I2C touch) from Elecrow's official schematic. Includes GPIO pin mappings and LVGL buffer initialization.

#### **LcdDisplay.h**

Public interface with methods: **begin()**, **loop()**, **showApMode()**, **updateDashboard()**

#### **LcdDisplay.cpp**

LVGL UI setup, AP-mode setup screen, live dashboard rendering directly from **IUSBHostUPS** data. Displays battery %, load %, runtime, real power, and color-coded UPS status (OL/OB/LB).

### 3. Add LVGL Conflict Fix Script

Create **scripts/fix\_lovyangfx\_lv\_conf\_conflict.py** to run pre-build:

- Remove stray **lv\_conf.h** copies from **.pio/libdeps/**
- Force **LV\_CONF\_PATH** to project's **include/lv\_conf.h** via absolute path

- Fixes LVGL's fragile **\_\_has\_include** auto-detection

*Why:* LovyanGFX and lvgl bundle their own **lv\_conf.h** inside **.pio/libdeps/** libraries, causing LVGL to silently miss required fonts (14px only, no 20px/32px). This script prevents that shadowing.

## 4. Update platformio.ini

Modify or create the **[env:esp32-s3-standard]** section to protect the generic build:

```
lib_ignore =  
    LcdDisplay
```

Create new **[env:elecrow-2-4-onboard]** section with:

```
platform = https://github.com/pioarduino/platform-espessif32.git#51.03.04  
board = esp32-s3-devkitc-1  
framework = arduino  
extra_scripts =  
    pre:scripts/embed_web.py  
    pre:scripts/fix_lovyangfx_lv_conf_conflict.py  
  
board_build.flash_size = 16MB  
board_build.f_flash = 80000000L  
board_build.psram_type = opi  
board_build.arduino.memory_type = qio_opi  
  
build_flags =  
    -DBOARD_HAS_LCD  
    -DBOARD_HAS_PSRAM  
    -DLV_CONF_INCLUDE_SIMPLE  
  
lib_deps =  
    ${env.lib_deps}  
    adafruit/Adafruit NeoPixel@^1.12.0  
    lvgl/lvgl@^8.3.11  
    lovyan03/LovyanGFX@^1.1.16
```

## 5. Modify src/core/main.cpp

Add guards and initialization for the LCD:

**At top, add global:**

```
#ifdef BOARD_HAS_LCD
String g_wifi_ssid;
#endif
```

**In setup(), after diagnostic\_led.begin():**

```
#ifdef BOARD_HAS_LCD
    lcd_display.begin();
#else
    diagnostic_led.begin(LED_BUILTIN_PIN);
#endif
```

**In setup(), in AP-mode branch:**

```
#ifdef BOARD_HAS_LCD
    lcd_display.showApMode(ap_ssid, ap_password);
#endif
```

**In loop(), update display every ~1 second:**

```
#ifdef BOARD_HAS_LCD
    static uint32_t last_lcd_update = 0;
    if (millis() - last_lcd_update >= 1000) {
        last_lcd_update = millis();
        lcd_display.updateDashboard(&usb_ups,
            network_mgr.isConnected(), g_wifi_ssid);
    }
#endif
```

*Note:* Skip **diagnostic\_led.begin()** entirely on LCD board—GPIO38 (NeoPixel power pin) conflicts with the LCD backlight pin. On-screen status dots replace the physical LED.

## 6. Modify include/core/main.h

Add LCD header guard:

```
#ifndef BOARD_HAS_LCD
#include "LcdDisplay.h"
extern LcdDisplay lcd_display;
#endif
```

## Build

```
pio run -e elecrow-2-4-onboard
```

The generic **esp32-s3-standard** environment remains unchanged and unaffected by all LCD additions.

*Generated 2026-09-18 from esp32-nut LCD integration session.*